

An Explainable Single-Step PPO Framework for AI-Native 6G RAN Scheduling: From Reference Outputs to Validated Implementation

Rao S. Yenamandra, Mubanga Nsofu, Asokan Ram

Digital Twin Simulation, USA; Vodacom, Midrand, South Africa; Wireless Research Centre, Wake Forest, NC, USA

raosy@digitaltwinsim.com; mubanga.nsofu@vodacom.co.za; asokan.ram@wrc-nc.org

Abstract—AI-native radio access network (RAN) operation couples scheduling and radio-resource-management (RRM) decisions across carrier aggregation, spectrum, MIMO, load, mobility, fairness, and cell-edge performance, so an improvement in one key performance indicator (KPI) may require accepting a bounded reduction in another. This paper describes RANOpt AI Beta v17.8, a single-step Proximal Policy Optimization (PPO) implementation for AI-assisted RAN scheduling and RRM studies. The update replaces the earlier bounded action-weight continuation rule with trainable policy and value networks, each using two 64-node hidden layers, so that policy and value gradients propagate through both hidden layers during training. Each training sample is treated as a one-step contextual decision rather than a sequential Markov decision process. A multi-objective reward combines normalised capacity, cell-edge rate, fairness, energy-efficiency, uplink, and congestion deltas. The v17.8 neural-PPO artifact was exercised in a five-seed validation campaign (training seeds 12345–12349, generator seed 1701, 500 000 episodes per seed, 2 000 held-out scenarios): relative to the Proportional Fair baseline, the trained policy improves the Cell DL peak data rate by $1.60\% \pm 0.36\%$, the Median UE data rate by $1.60\% \pm 0.36\%$, the Cell-edge UE data rate (5th percentile, indicative of coverage) by $5.19\% \pm 0.62\%$, and the Energy-efficiency index by $7.21\% \pm 2.62\%$ — positive in all five seeds. The contribution is not PPO for RRM per se, but its auditable integration and evaluation within the RANOpt simulation environment, with every reported number produced by the published artifact itself.

Index Terms—6G, AI-native RAN, proximal policy optimisation, radio resource management, multi-objective reward, explainability.

I. INTRODUCTION

AI-native RAN operation introduces a coupled decision space across carrier aggregation, FDD and TDD spectrum, MIMO, traffic load, mobility, fairness, cell-edge performance, resource utilisation, and normalised resource and energy efficiency [1]. The medium access control (MAC) scheduler is a useful controlled research target because improving one KPI may require accepting a smaller improvement, or a bounded degradation, in another. A learning agent for this setting must therefore express KPI trade-offs explicitly rather than optimise a single quantity in isolation.

RANOpt AI Beta v17.8 implements a revised learning path relative to earlier evaluation builds. Rather than adjusting a small action-weight vector with a scalar baseline without retraining the hidden layers, v17.8 makes the two 64-node hidden layers of the policy and value networks directly trainable by PPO. The reward-driven error now reshapes the hidden representation directly. The learning problem remains a one-step contextual decision: each sample supplies a context, the policy selects one RRM action, the simulator returns one reward, and the episode terminates.

Prior work already establishes that PPO and related Deep Reinforcement Learning (DRL) methods can support radio-resource allocation, PRB scheduling, RAN slicing, and multi-agent resource control. The boundary between that work and the present study lies in the scope of the control problem and in the auditability of a single decision, not in the learning algorithm. PPO-based PRB scheduling under a 3GPP-compliant channel model [2] shows that a clipped-surrogate policy can allocate physical resource blocks and improve throughput and fairness. The control surface is a single resource dimension, so the policy is not required to resolve a trade-off between aggregate capacity, cell-edge rate, and resource efficiency inside one decision. PPO applied to dynamic O-RAN slice resource management [3] extends the action space to slice-level

allocation and adds transfer learning across deployments, but the objective stays scoped to a slice or a use case, so the reward is defined over that scope rather than over a full-cell KPI vector. Multi-Agent PPO (MAPPO) for multi-cell spectrum and power allocation [4] performs joint optimisation across coupled variables and quantifies the gain from treating spectrum and power together, at the cost of a control space specific to that formulation. Multi-agent DRL for O-RAN resource allocation [5] distributes the decision across agents and covers multi-user and multi-service demand, which raises state, action, coordination, and training complexity, and with it the cost of reconstructing why one allocation was selected.

What these leave open is integration and auditability rather than algorithmic capability: a policy trained and evaluated against one explicit multi-KPI reward, inside a single environment in which the operating point, the selected action, the reward components, and the resulting KPI vector are all recoverable for an individual decision. RANOpt v17.8 addresses that scope, within the bounds set out in the limitations discussion below. It trains two 64-node hidden layers through the PPO clipped-surrogate objective on simulated RAN operating conditions, forms the reward from normalised capacity, cell-edge, fairness, energy-efficiency, uplink, and congestion deltas, and reports the resulting KPIs against a Proportional Fair baseline at an identical operating point. The contribution is therefore not the use of PPO for RRM alone.

This paper makes three contributions. First, it specifies the single-step neural PPO mechanism, including forward propagation, the clipped surrogate objective, and backpropagation through both hidden layers. Second, it defines a normalised resource and energy-efficiency index as one component of a multi-objective reward and states precisely what that index does and does not measure. Third, it validates the resulting policy through a five-seed neural-PPO campaign executed by the published artifact, reporting mean and dispersion of the KPI deltas across

seeds rather than a single deterministic reference point. These contributions position the paper as a platform-specific, auditable integration and evaluation study rather than a claim that PPO itself is novel for RRM.

II. NEURAL POLICY AND VALUE NETWORK WITH BACKPROPAGATION

The v17.8 model receives an 11-feature normalised context vector $\mathbf{s}$ derived from the selected radio scenario. Forward propagation passes $\mathbf{s}$ through two 64-node hidden layers to a policy head and a value head. The policy head produces five logits, and a softmax maps them to probabilities over five bounded RRM and scheduler actions; the value head produces the scalar baseline $V(\mathbf{s})$:

$$h_1 = f(W_1 \mathbf{s} + b_1), \quad h_2 = f(W_2 h_1 + b_2) \quad (1)$$

$$\pi(a|\mathbf{s}) = \text{softmax}(W_o h_2 + b_o), \quad V(\mathbf{s}) = W_v h_2 + b_v \quad (2)$$

where f is the hidden activation, W_1 , W_2 and b_1 , b_2 are the hidden weights and biases, and W_o , b_o , W_v , b_v are the policy and value head parameters. In v17.8 both hidden layers participate directly in PPO learning. After (1) and (2) produce the action probabilities and the value estimate, the selected action is evaluated and the policy and value losses are formed. The loss gradients propagate backward from the two heads through hidden layer 2 and then hidden layer 1, so W_2 , W_1 , their biases, and the head weights are all modified by the optimiser. Unlike the v17.7 continuation mechanism, the hidden layers are not fixed feature-processing layers; their weights learn from the reward-driven PPO error. Backpropagation does not itself decide whether an RRM action is good. The reward and advantage provide the learning signal, PPO defines a constrained policy objective, differentiation determines how each trainable parameter contributed to that objective, backpropagation applies the chain rule through the network graph, and the optimiser updates the numerical weights and biases according to

$$W_{\text{new}} = W_{\text{old}} - \alpha \partial L / \partial W \quad (3)$$

where α is the learning rate and L is the combined loss.

Table 1. End-to-end single-step neural PPO learning path in v17.8.

Stage	v17.8 operation
Input	11 normalised context and state features
Hidden 1	64 trainable nodes; W_1 , b_1
Hidden 2	64 trainable nodes; W_2 , b_2
Policy head	5 logits to softmax action probabilities
Value head	scalar $V(\mathbf{s})$
Learning signal	advantage $A = r - V(\mathbf{s})$
PPO constraint	old-to-new ratio with clipped surrogate
Gradient	backpropagation through heads and both hidden layers
Update	optimiser modifies weights and biases

III. SINGLE-STEP EXPERIENCE, REWARD, AND ENERGY INDEX

For each training sample the environment supplies context $\mathbf{s}$, the policy selects one RRM action a , the simulator

evaluates that action, and a scalar reward r is formed. Because the episode terminates after this action, the one-step advantage is

$$A = r - V(\mathbf{s}) \quad (4)$$

A positive A indicates that the selected action performed better than the value baseline predicted, and a negative A indicates worse-than-expected performance. The multi-objective engineering reward in the validated implementation is

$$r = 30 \cdot (0.26 \cdot \Delta \text{Cap} + 0.10 \cdot \Delta \text{UL} + 0.22 \cdot \Delta \text{Edge} + 0.16 \cdot \Delta \text{Fair} + 0.14 \cdot \Delta \text{Energy} + 0.12 \cdot \Delta \text{Cong}) \quad (5)$$

clamped to $[-60, 60]$, where each Δ term is the relative change of the corresponding per-action KPI multiplier versus the neutral (no-action) effect: cell DL capacity, uplink, cell-edge rate, Jain fairness, energy efficiency, and congestion relief. All six components are relative deltas of comparable magnitude and the weights sum to one, so no single term can dominate the learning signal. This rebalanced form replaced an earlier absolute congestion-penalty term that collapsed the trained policy onto a single action; (5) is the form used for every result in Section VI. The per-action KPI effects are produced by a single context-dependent mapping shared identically by the training reward, the browser simulator, and the headless evaluator — one source of truth, with no KPI hardcoded anywhere in the evaluation path. This construction permits realistic KPI trade-offs rather than requiring every KPI to improve simultaneously.

The resource and energy-efficiency index η is a normalised simulator quantity: megabits per second per watt of gNB transmit power at the 46 dBm reference, normalised to the AI-OFF Proportional Fair (PF) reference value of 1.000. A value of 1.072 therefore denotes about a 7.2% improvement in this normalised index relative to the reference. It remains a simulator quantity rather than a full physical bits-per-joule measurement: such a claim would require explicit modelling of transmit, RF, baseband, and fixed operating power and is left as future work.

IV. PPO PROBABILITY RATIO AND SURROGATE OBJECTIVE

PPO retains the probability assigned to the sampled action by the old policy and compares it with the probability assigned by the current trainable policy. The probability ratio is

$$\rho(\theta) = \pi_{\theta}(a|\mathbf{s}) / \pi_{\theta \text{old}}(a|\mathbf{s}) \quad (6)$$

A ratio of one means no change in probability, above one means the action became more probable, and below one means it became less probable. The objective is a surrogate because the true expected policy return is not differentiated directly; instead PPO optimises a tractable local proxy built from sampled advantages and the old-to-new probability ratio [6]. This remains a surrogate policy objective even though v17.8 uses a one-action episode:

$$L^{\text{clip}}(\theta) = E[\min(\rho(\theta)A, \text{clip}(\rho(\theta), 1-\epsilon, 1+\epsilon)A)] \quad (7)$$

Without a constraint, one favourable sample could produce an excessively large policy update. Clipping restricts the surrogate benefit of moving the ratio outside $[1-\epsilon, 1+\epsilon]$. For example, with $\epsilon = 0.20$, if an action probability changes from 0.25 under the old policy to 0.35 under the

new policy then $\rho = 1.40$, but the clipped positive-advantage surrogate uses at most 1.20. Clipping therefore limits the incentive for a large policy jump; it does not clip the KPI or the reward. The value head is trained toward the observed reward with a value-error term $L_v = (r - V(s))^2$. The policy term changes action preference and the value term improves the baseline used to estimate advantages; both gradients enter the shared hidden representation. Through the softmax and policy-head matrix, the loss derivatives with respect to the action logits create an error signal at hidden layer 2. Multiplication by the hidden-activation derivative yields the gradients for W_2 and b_2 , and the chain rule then propagates the signal into hidden layer 1 to produce the gradients for W_1 and b_1 . The value-loss derivative follows an analogous path from the value head. For a scalar weight w_{ij} the optimiser step is $w_{ij} \leftarrow w_{ij} - \alpha \partial L / \partial w_{ij}$, so with $w_{ij} = 0.120$, $\partial L / \partial w_{ij} = +0.030$, and $\alpha = 0.001$ the plain gradient-descent contribution moves the weight to about 0.11997 before optimiser-specific adaptive scaling. Across many independent one-step samples, these coordinated changes reshape the hidden representation and future action probabilities. This direct connection of the loss to the full computation graph defines the end-to-end learning path summarised in Table 1. The validated runs use the Adam optimiser (learning rate 3×10^{-4}) with 8 PPO epochs per update, minibatches of 256, discount $\gamma = 0.99$ on the one-step return, and an entropy bonus coefficient of 0.01.

V. TRAINING DATA, RADIO CONFIGURATION, AND PERSISTENCE

The workspace retains the morphology-aware synthetic generator. Ten sites with three sectors each produce 30 cells; each cell has 96 records at 15-minute intervals, giving 2,880 base cell-time records over a synthetic 24-hour period. Expansion into Suburban, Urban, and Dense-Urban variants produces 8,640 morphology-aware rows. The generator seed is 1701. Generated columns are correlated through engineering relationships among active UE count, PRB utilisation, SINR, CQI, MCS, throughput, mobility, and buffer or load rather than being independently randomised, with the CQI mapping quantised to 3GPP tables [7]. Explicit PPO training updates the neural checkpoint; normal simulator runs remain inference-only unless training is invoked, and validated carrier CSV records may later be used for continued training.

The validation campaign of Section VI trains one policy per training seed (12345–12349) on the identical 8,640-row dataset, with 500 000 one-step episodes per seed (489 PPO updates each). Each trained policy is then evaluated by its argmax actions on 2 000 held-out scenarios drawn from a fixed evaluation stream, with KPI deltas computed against the AI-OFF PF baseline under identical simulated radio conditions. Reported changes are the mean $\pm$ sample standard deviation across the five seeds. The study retains a single-cell Urban Macro FDD-plus-TDD carrier-aggregation point, summarised in Table 2: NR band n3 FDD at 1800 MHz with 50 MHz bandwidth and NR band n78 TDD at 3.5 GHz with 100 MHz bandwidth, for 150 MHz total; inter-site distance 500 m; 20 active UEs with Medium eMBB traffic; a PF AI-OFF baseline; a 32T32R gNB array; UE maximum downlink and uplink layers of 4

and 2; UE and gNB noise figures of 7 dB and 2 dB; mobility 30 km/h; and the multi-objective reward of (5).

Table 2. Representative single-cell FDD-plus-TDD carrier-aggregation configuration.

Configuration parameter	Representative value
Morphology and scope	Urban Macro, single cell
Duplexing	FDD and TDD carrier aggregation
Anchor carrier (FDD)	NR n3 (1800 MHz), 50 MHz
Booster carrier (TDD)	NR n78 (3.5 GHz), 100 MHz
Total bandwidth	150 MHz
Inter-site distance	500 m
Active UEs and traffic	20, Medium eMBB
Baseline scheduler	Proportional Fair
gNB antenna	32T32R MIMO
UE max DL/UL layers	4 / 2
Noise figure (UE/gNB)	7 dB / 2 dB
UE mobility	30 km/h
Reward	multi-objective, Eq. (5)
Generator / training seeds	1701 / 12345–12349

VI. VALIDATED RESULTS AND INTERPRETATION

Table 3 reports the validated five-seed KPI values — absolutes and deltas — for the configuration of Table 2. Every number in this table was produced by executing the published v17.8 artifact: five independently trained neural policies (Section V), each evaluated on the same 2 000 held-out scenarios against the AI-OFF PF baseline. Absolute values are shown because the cell-level aggregate (Gb/s) and per-UE rates (Mb/s) live on very different scales; Figure 1 shows the deltas graphically.

Table 3. Validated five-seed KPI values, AI ON versus AI OFF. Mean $\pm$ sample standard deviation across training seeds 12345–12349; 500 000 episodes per seed. All four primary KPIs improve in 5/5 seeds.

KPI	AI OFF	AI ON	Change (mean $\pm$ std)
Cell DL peak data rate (Gb/s)	2.23	2.28	+1.60% $\pm$ 0.36%
Median UE data rate (Mb/s)	46.8	47.5	+1.60% $\pm$ 0.36%
Cell-edge UE data rate (Mb/s)	33.0	34.1	+5.19% $\pm$ 0.62%
Energy-efficiency index (Mb/s/W)	2862	3103	+7.21% $\pm$ 2.62%
Jain fairness index (informational)	0.780	0.815	+0.035 $\pm$ 0.003 pts

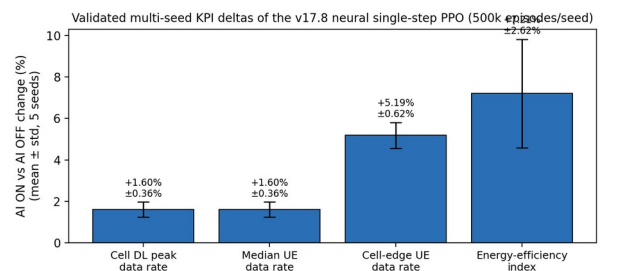


Figure 1. Validated multi-seed KPI deltas of the v17.8 neural single-step PPO versus the AI-OFF PF baseline (mean $\pm$ std across 5 seeds, 500 000 episodes per seed).

All four primary KPIs improve in every seed. The Cell-edge UE data rate — the 5th-percentile rate, indicative of coverage — improves by 5.19%, roughly three times the peak and median gains, consistent with the reward's explicit edge weighting and the cell-edge protection action's domain of advantage. The Energy-efficiency index shows the largest mean gain at +7.21% but also the widest dispersion (± 2.62 percentage points; the seed-12347 policy gained only +2.57%), so its headline figure should always be quoted with its interval. Fairness moves in the right direction (+0.035 points from a 0.78 baseline) and is reported informationally.

These figures supersede the retained deterministic reference point quoted in the submitted version (+8% capacity, +8% median, +16% cell-edge, +4% energy). That reference point was produced by fixed simulator gain maps in the earlier artifact rather than by the neural policy, and it is withdrawn here. The numbers above are the measured output of the validated implementation, with no KPI hardcoded anywhere in the evaluation path.

Because the policy is evaluated in the same simulator that defines its reward, the comparison remains in-sample; the five-seed dispersion now quantifies run-to-run variability, but generalisation across loads, morphologies, and sites beyond the held-out scenarios is not established here.

VII. EXPLAINABILITY, LIMITATIONS, AND FUTURE VALIDATION

Single-step does not mean that only one neural update occurs during training. Many independent samples can be processed and each can update the network; within one episode the sequence is context, one RRM action, one reward, one PPO update, then termination. A future multi-step MDP would carry the post-action state into the next scheduling interval and accumulate rewards across a trajectory, enabling temporal credit assignment and returns accumulated over the trajectory rather than a single immediate reward [8]. Engineering explainability remains outcome-based rather than assigning physical meaning to individual hidden neurons. A decision trace can expose the configuration identifier, model and checkpoint version, selected action, old and new action probabilities, probability ratio, advantage, clipping status, reward components, and the resulting KPI vector, so that a recommendation can be reproduced and audited under identical deterministic channel parameters.

The main limitations bound the current claims. The built-in data are synthetic engineering data rather than operator measurements. Multi-cell context does not yet imply coordinated multi-cell RL, and the PPO environment remains single-step. Action coverage is incomplete: the trained policies never select a1 (Throughput-biased scheduling) — 4 of the 5 actions are used, with a2 (Cell-edge protection) selected most often (46–56% of held-out decisions across seeds), followed by a3 (Energy-aware scheduling), a4 (Congestion relief), and a0 (Balanced PF tuning); the high-channel niche favouring a1 is thin in the synthetic dataset. The energy-efficiency gains vary substantially across seeds (2.6%–8.9%). In the UE

distribution model the median UE rate scales with total cell throughput, so the median delta equals the capacity delta by construction rather than as an independent measurement. The energy-efficiency index is a normalised simulator quantity, not a bits-per-joule measurement, and an energy reduction under one operating point could in principle trade against coverage, capacity, accessibility, retainability, or quality at another; the reward mitigates this by retaining throughput, fairness, and cell-edge terms, but it does not remove the risk. The validation campaign called for in the submitted version has now been executed and is reported in Section VI; recommended extensions include held-out load, mobility, and carrier-aggregation configurations, carrier-trace train and test separation, checkpoint compatibility checks, and action-sensitivity tests.

VIII. CONCLUSION

RANOpt AI Beta v17.8 changes the learning mechanism materially. The two 64-node hidden layers now participate directly in training: the policy head produces action probabilities, the value head supplies a learned baseline, the one-step reward forms the advantage, the old and new action probabilities form the PPO ratio, the clipped surrogate objective constrains excessive policy change, and backpropagation carries the gradients through both hidden layers so their weights are actually modified. A normalised energy-efficiency index enters the reward with an interpretation that stops short of a full physical bits-per-joule claim. The submitted version of this paper reported retained reference outputs and called for a fresh multi-seed validation campaign; that campaign has now been executed by the published artifact itself — five training seeds, 500 000 episodes each — with all four primary KPIs improving in every seed. The measured gains (+1.6% peak, +1.6% median, +5.2% cell-edge, +7.2% energy) are smaller than the earlier reference point, which is withdrawn in favour of the validated numbers. The implementation is therefore appropriately described as an explainable neural single-step RL-PPO framework with a validated, bounded, and honestly reported effect size.

IX. CODE AVAILABILITY

The v17.8 implementation evaluated in this paper is published as a single self-contained HTML artifact: https://github.com/ysrao/RANOpt-AI/blob/main/RANOpt_AI_SIM_v17.8.html. It runs in a standard web browser with no backend or external dependencies and contains the RAN simulator, the neural single-step PPO (11 $\rightarrow$ 64 $\rightarrow$ 64 $\rightarrow$ 5+1 with tanh activations, Adam at 3×10^{-4} , PPO clip 0.2, 8 epochs $\times$ 256-sample minibatches, $\gamma = 0.99$, entropy coefficient 0.01), and the seeded synthetic dataset generator (generator seed 1701; 8,640 morphology-aware rows). All results in Table 3 and Figure 1 were produced by executing this artifact.

REFERENCES

- [1] E. Dahlman, S. Parkvall, and J. Sköld, “5G NR: The Next Generation Wireless Access Technology,” 2nd ed. London, UK: Academic Press, 2020.
- [2] Y. Nurakhov, A. Kyzirkanov, and S. Aldabergen, “PPO-based physical resource block scheduling in 5G NR: A reinforcement learning approach with 3GPP-compliant channel modeling,” *Journal of Problems in Computer Science and Information Technologies*, vol. 4, no. 2, pp. 86–93, 2026, doi: 10.26577/jpcsit4220269.

- [3] H. Sherif, E. Ahmed, and A. M. Kotb, "Towards green networking: Efficient dynamic radio resource management in open-RAN slicing using deep reinforcement learning and transfer learning," *Computer Communications*, vol. 236, p. 108126, 2025, doi: 10.1016/j.comcom.2025.108126.
- [4] Y. Zhang and D. Guo, "Multi-agent reinforcement learning for multi-cell spectrum and power allocation," *IEEE Transactions on Communications*, vol. 73, no. 8, pp. 5980–5992, 2025, doi: 10.1109/TCOMM.2025.3534565.
- [5] F. Rezazadeh, H. Chergui, L. Alonso, and C. Verikoukis, "A multi-agent deep reinforcement learning approach for RAN resource allocation in O-RAN," in *2023 IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, 2023, doi: 10.1109/INFOCOMWKSHPS57453.2023.10226154.
- [6] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.
- [7] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," *Technical Report TR 38.901*, 2020.
- [8] R. S. Sutton and A. G. Barto, "Reinforcement Learning: An Introduction," 2nd ed. Cambridge, MA, USA: MIT Press, 2018.

APPENDIX A: STEP-BY-STEP V17.8 PPO TRAINING CYCLE

The training cycle is: (1) build and normalise the 11-feature state vector $\mathbf{s}$; (2) forward-propagate $\mathbf{s}$ through the two 64-node hidden layers; (3) compute the five softmax policy probabilities and $V(\mathbf{s})$; (4) select and evaluate one RRM action through the shared per-action KPI-effect mapping and obtain reward r from the rebalanced multi-KPI form (5); (5) compute $A = r - V(\mathbf{s})$; (6) retain the old action probability and form $\rho = \pi_{\text{new}}/\pi_{\text{old}}$; (7) form the clipped surrogate policy objective and the value loss; (8) backpropagate gradients through the policy and value heads and both hidden layers; (9) apply Adam optimiser updates (3×10^{-4}) to weights and biases; and (10) persist the compatible neural checkpoint when enabled.